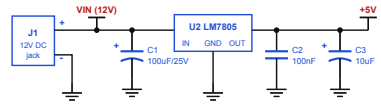
PIC16F877A Temperature & Humidity Monitoring System - Schematic

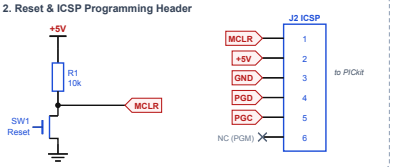
Net labels with the same name are electrically connected. All GND symbols are one common ground.

1. Power Supply (12 V in -> 5 V regulated)

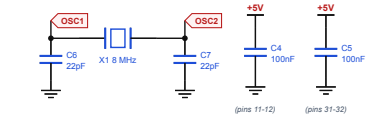


Use a 12 V / 1 A adapter. Fit a small heatsink on the LM7805.

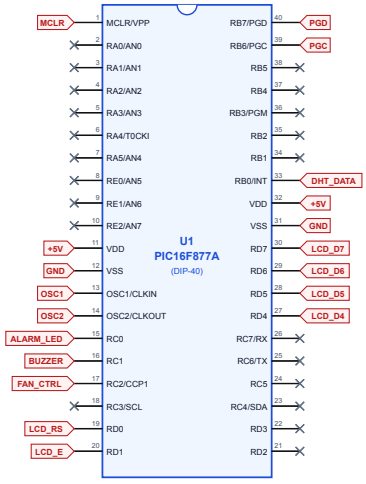
2. Reset & ICSP Programming Header



3. Clock (8 MHz) & Decoupling

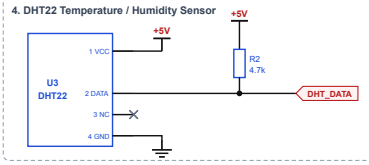


Place C4/C5 right at the chip

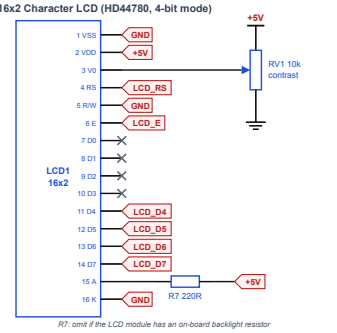


X = unused pin (leave open, set as output low in code)

4. DHT22 Temperature / Humidity Sensor

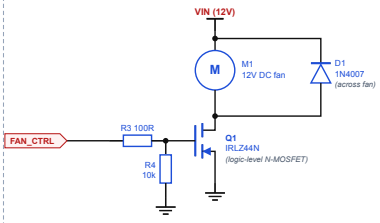


5. 16x2 Character LCD (HD44780, 4-bit mode)

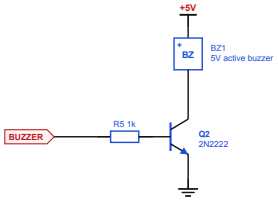


R7: omit if the LCD module has an on-board backlight resistor

6. Fan Driver (main actuator)



7. Buzzer Alarm (humidity / over-temperature)



8. Alarm LED & Firmware Notes



Configuration bits (XC8 #pragma config):
FOSC = HS, WDTE = OFF, PWRTE = ON, BOREN = ON,
LVP = OFF (frees RB3), CPD = OFF, WRT = OFF, CP = OFF

Pin summary:
RB0 = DHT22 data RD0/RD1 = LCD RS/E
RD4-RD7 = LCD D4-D7 RC2 = fan (CCP1, PWM-capable)
RC1 = buzzer RC0 = alarm LED RB6/RB7 = ICSP

Control: fan ON at T >= 30 C, OFF at T <= 28 C;
buzzer + LED ON at RH >= 80 %.